

ECE531 Computer Vision — Assignment 1

Low-Level Vision: Enhancement, Filtering, Geometry, and Local Structure

Student Name: İlayda Dinçkūlah

Student ID: 2540041006

Date: 29.03.2026

1. Introduction

This report presents a complete low-level vision pipeline applied to a custom dataset of fourteen photographs. The pipeline includes point-wise enhancement, spatial filtering, frequency-domain processing, edge detection, corner detection, and planar rectification. The goal is to compare classical computer vision methods and discuss their strengths under different imaging conditions such as low light, backlighting, repeated structures, shallow depth-of-field, and strong perspective.

2. Dataset Description

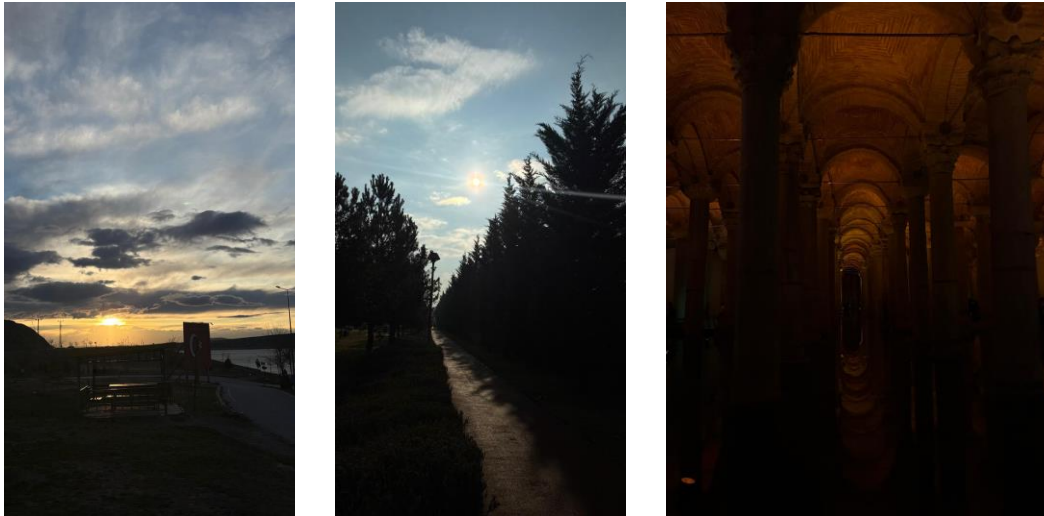
The dataset contains fourteen images selected to satisfy the assignment requirements. They include: (i) a building facade with repeated structural elements and perspective, (ii) a very dark interior image, (iii) a backlit outdoor pathway, (iv) a sunset scene with silhouettes, (v) a close-up document image with blur and handwriting, (vi) a planar sign suitable for homography-based rectification, and (vii) a corridor image with strong vanishing lines.

These images were intentionally selected to cover different low-level vision scenarios. For example, the dark interior and backlit pathway are useful for enhancement experiments, while the building facade and corridor are effective for edge and corner analysis due to their strong geometric structure. The planar sign image is used for projective rectification because it contains a nearly rectangular planar surface under perspective distortion.

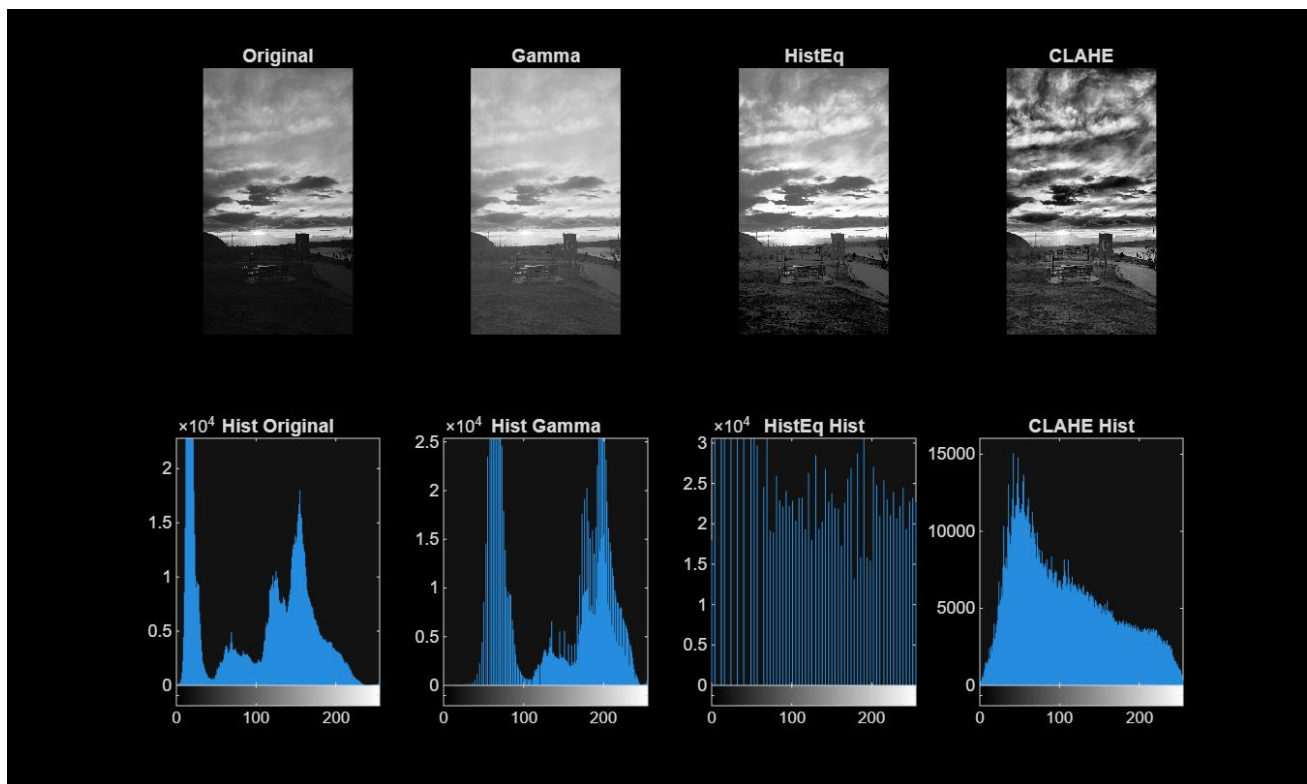
3. Task 1 - Point Operations and Intensity Enhancement

Methods used: Gamma correction, Histogram Equalization, and CLAHE.

Original Images:



Results:



TASK 1 Results - wp2.jpeg

Original



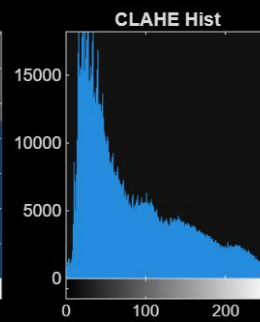
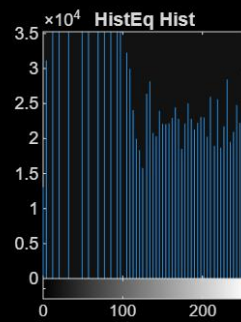
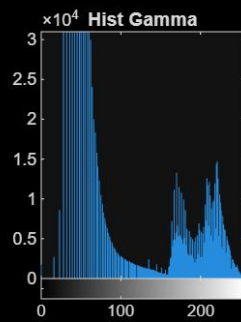
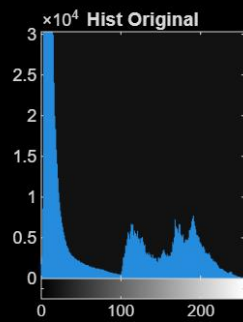
Gamma



HistEq



CLAHE



TASK 1 Results - wp3.jpeg

Original



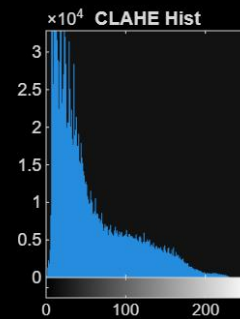
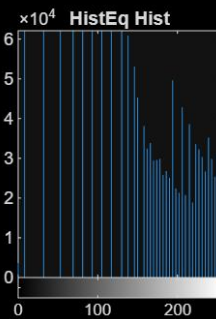
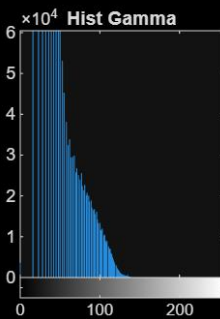
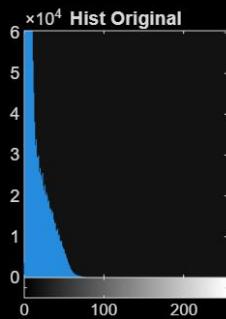
Gamma



HistEq



CLAHE



In this task, three point-wise intensity enhancement methods—gamma correction, histogram equalization, and CLAHE—were applied to images with challenging lighting conditions such as low-light interiors and backlit outdoor scenes. The goal was to improve visibility, contrast, and overall image quality while analyzing their effects using both visual results and histogram distributions.

Gamma correction improved the visibility of darker regions by redistributing intensity values toward higher levels, making shadowed areas more distinguishable without significantly affecting bright regions. This is clearly observed in the histograms, where pixel values shift toward mid-intensity ranges. However, the overall contrast improvement remained moderate.

Histogram equalization significantly enhanced global contrast by spreading pixel intensities across the entire dynamic range. The corresponding histograms became more uniformly distributed, indicating effective contrast stretching. While this improved visibility in many regions, it also led to over-enhancement in bright areas and loss of natural appearance, particularly in sky regions and highlights.

CLAHE (Contrast Limited Adaptive Histogram Equalization) provided the most balanced results among the three methods. It enhanced local contrast by applying histogram equalization within small regions while limiting noise amplification. As seen in the results, CLAHE preserved structural details and improved visibility in both dark and mid-tone areas without introducing excessive brightness. The histograms show a well-distributed intensity range while maintaining smoother transitions compared to global histogram equalization.

Overall, CLAHE produced the most visually pleasing and informative results, especially for low-light and high dynamic range images. Gamma correction provided controlled brightness enhancement, while histogram equalization offered strong but sometimes excessive contrast improvement. These results demonstrate that local adaptive methods are more effective for handling complex illumination conditions.

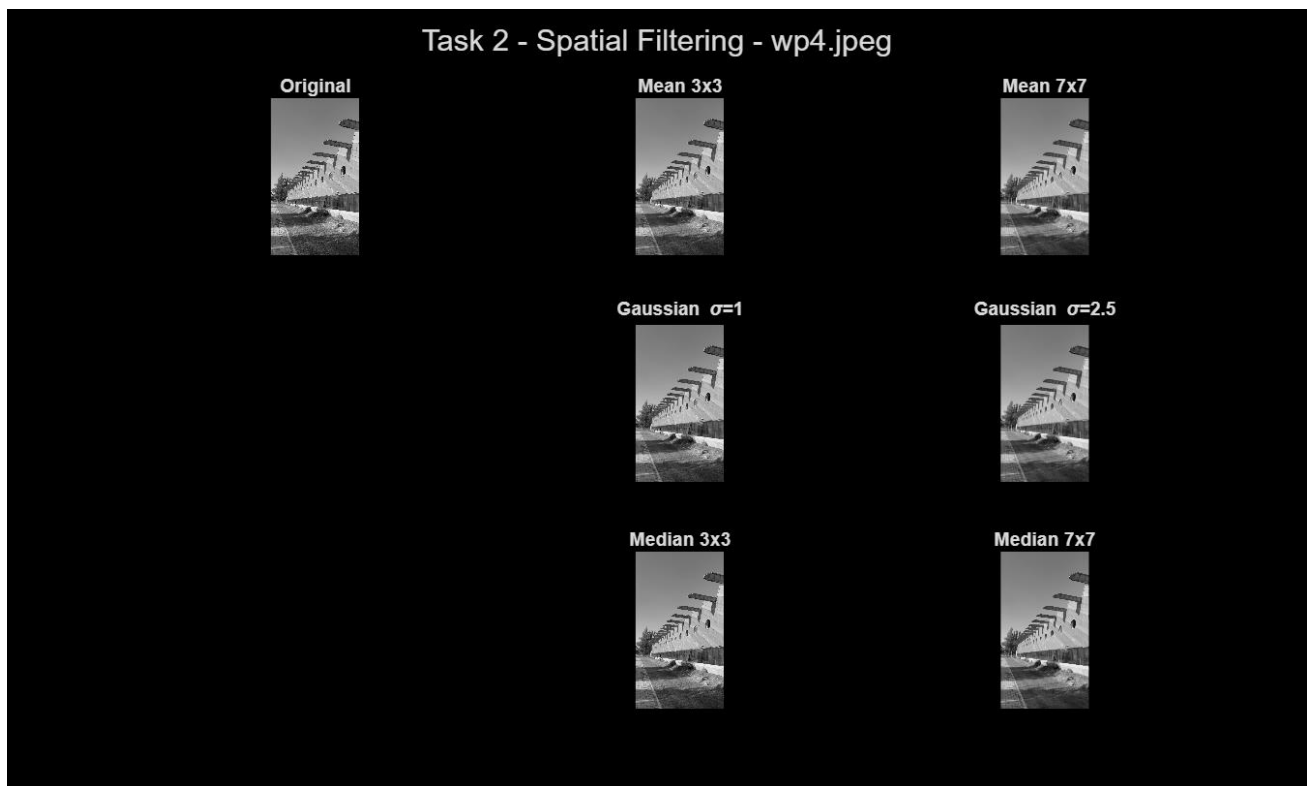
4. Task 2 - Spatial Filtering

Methods used: Mean, Gaussian, Median

Original Images



Results



Task 2 - Spatial Filtering - wp5.jpeg

Original



Mean 3x3



Mean 7x7



Gaussian $\sigma=1$



Gaussian $\sigma=2.5$



Median 3x3



Median 7x7



Task 2 - Spatial Filtering - wp6.jpeg

Original



Mean 3x3



Mean 7x7



Gaussian $\sigma=1$



Gaussian $\sigma=2.5$



Median 3x3



Median 7x7



In this task, spatial filtering techniques were applied using different kernel sizes to analyze the trade-off between smoothing and detail preservation. Three filters were evaluated: mean, Gaussian, and median filtering. For each method, two different configurations were tested to satisfy the requirement of using multiple kernel sizes. Specifically, mean filtering was applied with 3×3 and 7×7 kernels, Gaussian filtering was performed with $\sigma = 1$ and $\sigma = 2.5$, and median filtering was tested with 3×3 and 7×7 neighborhoods.

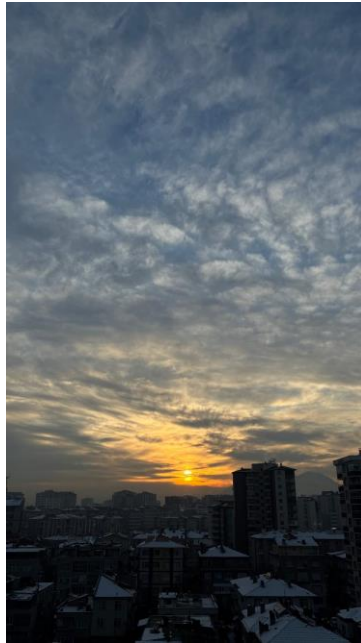
The results clearly demonstrate that increasing the kernel size or smoothing parameter leads to stronger noise reduction but also causes noticeable loss of fine structural details. In the corridor image (wp5), larger kernels such as mean 7×7 and Gaussian $\sigma=2.5$ significantly smoothed the repeating column patterns and shadow boundaries, reducing visual sharpness. In contrast, smaller kernels (3×3 or $\sigma=1$) preserved these linear structures more effectively. Similarly, in the sign image (wp6), larger filters caused slight blurring around text edges, while the median 3×3 filter maintained sharper character boundaries compared to the other methods. In the building façade image (wp4), Gaussian filtering provided a more balanced result by reducing intensity variations while preserving the geometric structure better than the mean filter.

When comparing denoising quality and edge preservation, mean filtering showed the strongest smoothing effect but also resulted in the greatest loss of detail. Gaussian filtering achieved a better balance between smoothing and structure retention, while median filtering was the most effective at preserving edges and sharp transitions, particularly in text regions. Overall, these results confirm that kernel size selection plays a critical role in spatial filtering: larger kernels improve smoothing but degrade details, whereas smaller kernels better preserve structural information. Therefore, the choice of filter and kernel size should be guided by the specific requirements of the application.

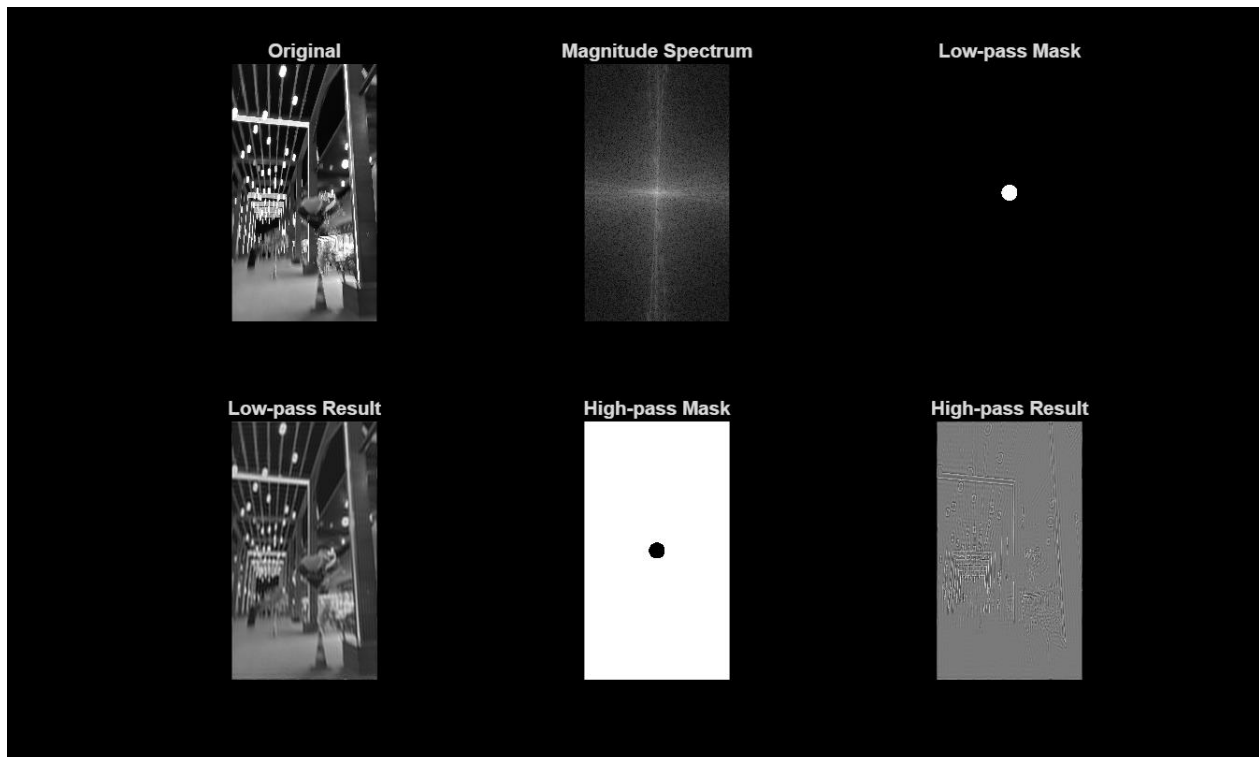
5. Task 3 - Frequency-Domain Processing

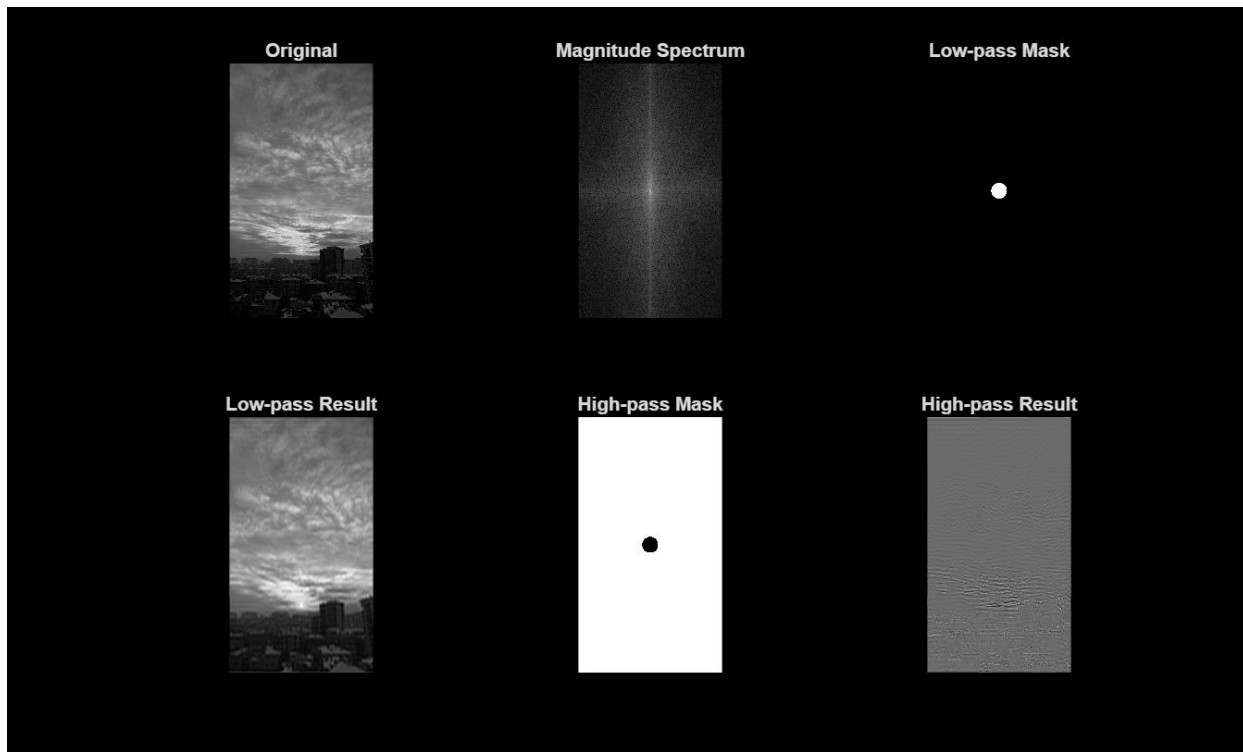
Methods used: FFT + Low-pass, High-pass

Original Images



Results





In this task, two images were transformed into the frequency domain using the Fourier Transform in order to analyze and manipulate their frequency components. The magnitude spectrum of each image was visualized after applying a logarithmic transformation, revealing that most of the image energy is concentrated around the center of the spectrum, corresponding to low-frequency components, while high-frequency components are distributed toward the outer regions.

Low-pass filtering was implemented by applying a circular mask centered at the origin of the frequency spectrum. This operation preserved low-frequency components while removing high-frequency details, resulting in smoother and more blurred images. This effect is clearly visible in both images: in the motion-blurred scene, the low-pass result further smooths the image and removes remaining fine variations, while in the urban landscape image, the low-pass filter suppresses structural details such as building edges while preserving the smooth sky regions.

High-pass filtering was performed by removing the low-frequency components and retaining high-frequency information. The resulting images emphasize edges and rapid intensity changes. In the first image, high-pass filtering highlights the structural outlines of lights and vertical elements, while suppressing the already blurred background. In the second image, high-pass filtering clearly extracts building boundaries and texture variations while removing the smooth intensity transitions of the sky.

These results demonstrate the direct relationship between frequency content and image structure: low frequencies correspond to global illumination and smooth regions, whereas high frequencies represent edges, textures, and fine details. Compared to spatial-domain filtering, frequency-domain filtering provides more explicit control over which frequency components are preserved or removed. While spatial filters such as Gaussian and mean filters achieve smoothing indirectly, low-pass filtering in the frequency domain performs a similar operation by directly attenuating high-frequency components. Similarly, high-pass filtering corresponds to edge-enhancing operations in the spatial domain. Overall, frequency-domain processing offers a more flexible and interpretable framework for analyzing and manipulating image content.

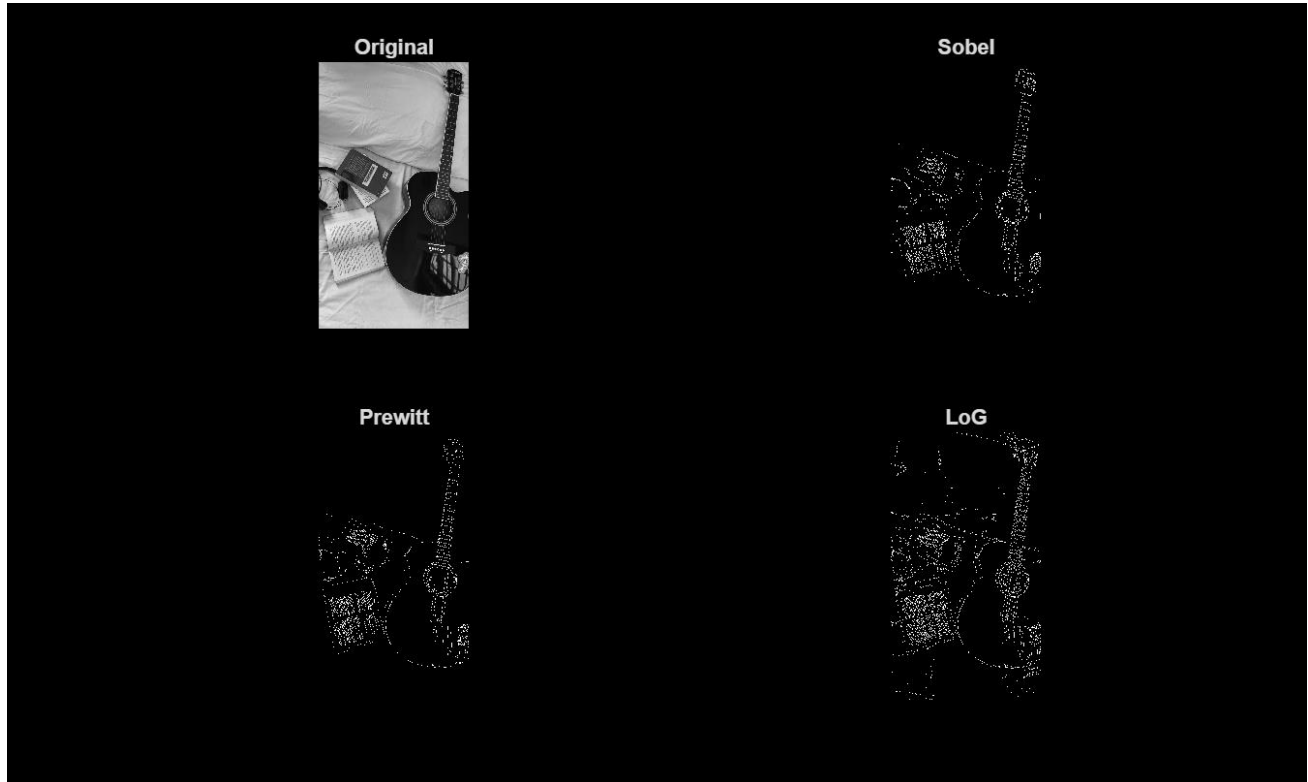
6. Task 4 - Edge Detection

Methods used: Sobel, Prewitt, and Laplacian of Gaussian.

Original Images



Results



Original



Sobel



Prewitt



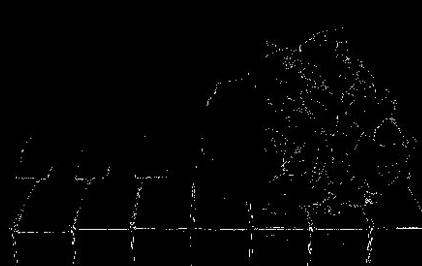
LoG



Original



Sobel



Prewitt



LoG



In this task, three edge detection methods—Sobel, Prewitt, and Laplacian of Gaussian (LoG)—were applied to multiple images to analyze their performance in terms of edge continuity, edge thickness, and sensitivity to noise. All methods were applied on grayscale images to ensure consistency in intensity-based edge detection.

The Sobel operator produced relatively continuous edges and was effective at capturing dominant structural boundaries, such as the contours of objects and strong intensity transitions. However, the detected edges appeared slightly thicker due to the gradient-based nature of the operator. The Prewitt method showed similar behavior but resulted in slightly weaker and less continuous edges compared to Sobel, particularly in low-contrast regions. This indicates that Sobel provides better edge continuity and robustness.

The Laplacian of Gaussian (LoG) method, which combines Gaussian smoothing with second-order derivative edge detection, was more sensitive to fine details and noise. While it was able to capture a larger number of edges, including subtle textures such as the flower petals and surface details of objects, it also introduced more noise-like responses. As a result, the edges detected by LoG appeared less continuous and more fragmented compared to Sobel and Prewitt.

In terms of noise sensitivity, Sobel and Prewitt were more stable, whereas LoG showed higher sensitivity due to its second-order derivative nature. Regarding edge thickness, Sobel and Prewitt produced thicker edges, while LoG resulted in thinner but less stable edge structures.

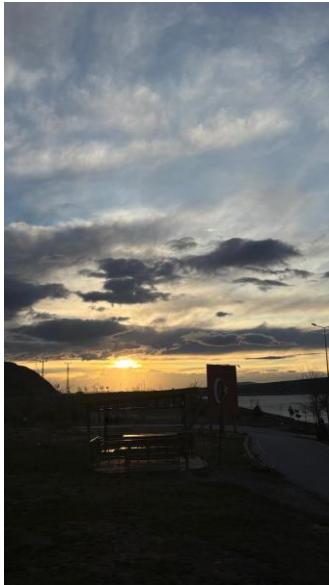
A notable failure case was observed in low-light and low-contrast regions, particularly in the dark violin image, where all methods struggled to detect clear edges due to insufficient intensity variation. Additionally, LoG produced excessive responses in textured areas, such as the flower and fabric regions, highlighting its sensitivity to noise and fine detail. Parameter choices, such as the implicit smoothing in LoG and the gradient thresholds in Sobel and Prewitt, significantly influenced the results, emphasizing the importance of proper parameter tuning in edge detection tasks.

Overall, Sobel provided the best balance between edge continuity and robustness, Prewitt offered a simpler but slightly weaker alternative, and LoG was more suitable for detecting fine details at the cost of increased noise sensitivity.

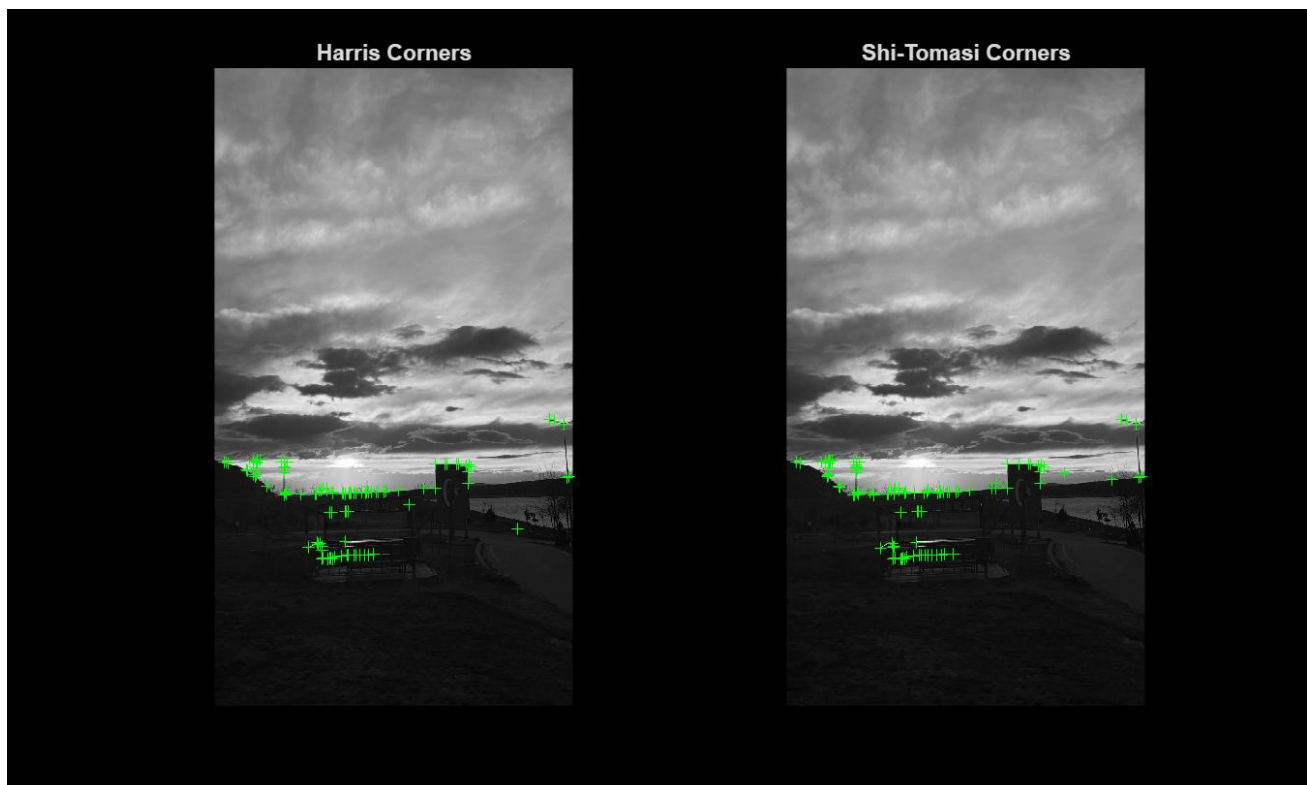
7. Task 5 - Corner / Interest Point Detection

Methods used: Harris and Shi-Tomasi.

Original Images



Results



Harris Corners



Shi-Tomasi Corners



Harris Corners



Shi-Tomasi Corners



In this task, corner detection was performed using two different methods: the Harris corner detector and the Shi–Tomasi method. Both methods were applied to multiple grayscale images, and the detected interest points were visualized on the images. The results show that both algorithms successfully identify corner-like structures, particularly in regions with strong intensity variations such as object boundaries, text edges, and structural details in buildings.

The Harris detector tends to produce a larger number of corner points, including some weaker and noisier detections, especially in textured regions such as clouds or grass. In contrast, the Shi–Tomasi method provides more stable and reliable results by selecting corners based on the minimum eigenvalue criterion, resulting in fewer but more distinctive points. This makes Shi–Tomasi more robust to noise and better suited for applications requiring precise feature selection.

In terms of sensitivity to texture and scale, both methods respond strongly to highly textured areas, which may lead to dense clusters of detected points. However, flat regions with little intensity variation produce almost no detections, as expected. When the scale of structures increases, both methods still detect corners effectively, but parameter tuning (e.g., threshold) becomes important to avoid excessive or insufficient detections.

From a conceptual perspective, corners differ from edges and flat regions in that they exhibit significant intensity changes in multiple directions. While edges represent intensity variation in only one direction, corners contain strong gradients in both horizontal and vertical directions, making them more informative and stable for feature-based tasks such as matching and tracking.

Overall, the comparison demonstrates that while Harris is more sensitive and detects more points, Shi–Tomasi provides better quality and stability of detected corners, making it a preferable choice in many practical computer vision applications.

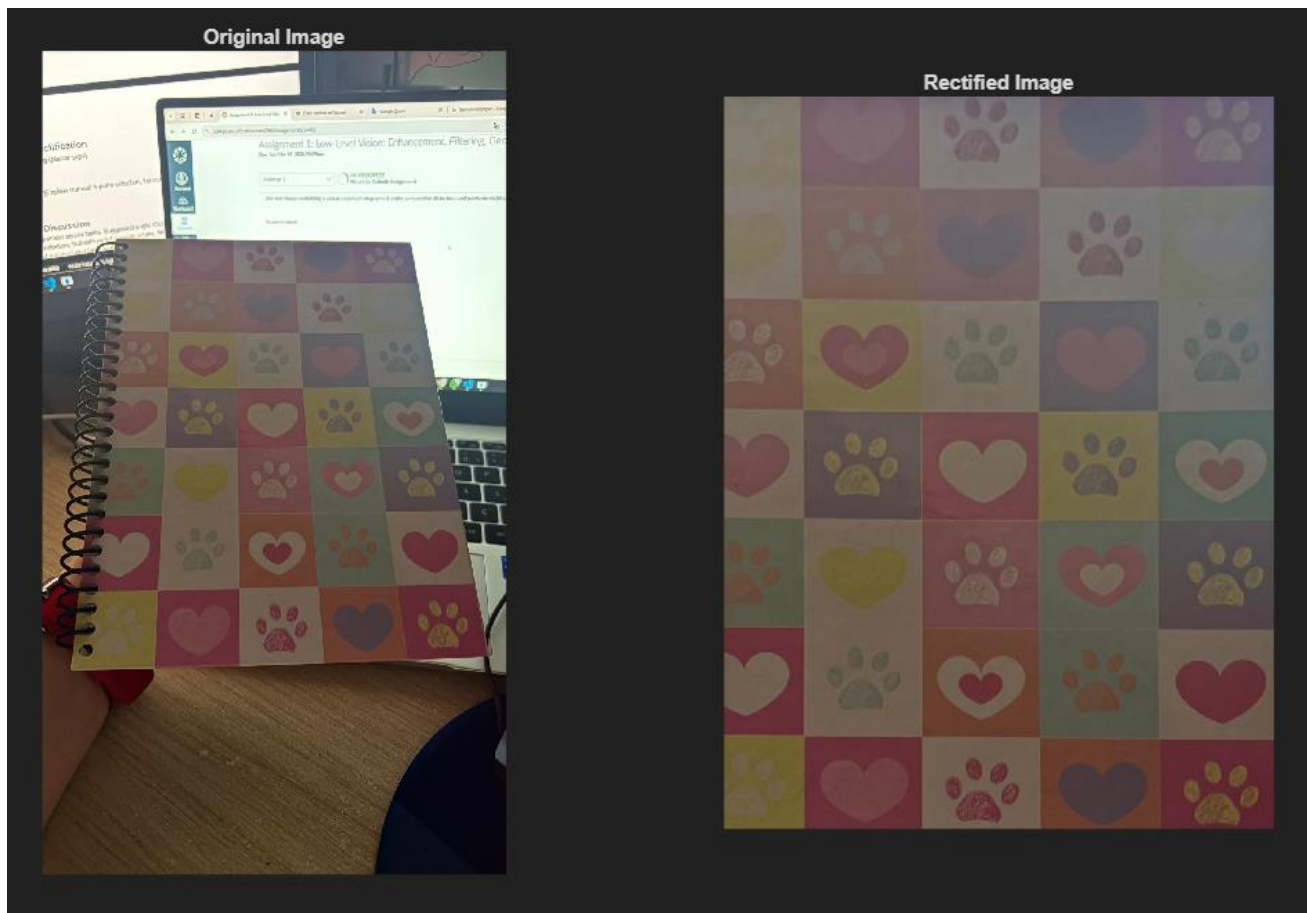
8. Task 6 - Planar Rectification

Rectification image: wp15.jpeg

Original Image



Result



In Task 6, planar rectification was performed on an image containing a notebook photographed under perspective distortion. The front surface of the notebook was selected as the planar object, and four corner points were manually identified in the original image. These correspondences were mapped onto a rectangular fronto-parallel plane, and a projective transformation (homography) was estimated to remove the perspective effect.

The rectified result shows that the notebook surface becomes geometrically more regular and closer to a frontal view after transformation. This makes the planar structure easier to analyze and visually confirms the effectiveness of projective rectification. The method assumes that the selected object is planar and that the corner points are chosen accurately. Small inaccuracies in point selection may slightly distort the final rectified image. Nevertheless, the result demonstrates how projective geometry can be used to recover the frontal appearance of a planar surface from a perspective image.

9. Comparative Discussion

Across all tasks, different classical computer vision methods were evaluated under varying imaging conditions, revealing clear trade-offs between enhancement, smoothing, edge detection, and feature extraction techniques.

In Task 1, CLAHE provided the most balanced enhancement results, particularly in low-light images, as it improved local contrast without over-amplifying noise. Compared to global histogram equalization, CLAHE preserved visual realism more effectively.

In Task 2, spatial filtering results showed that Gaussian filtering achieved the best compromise between noise reduction and detail preservation, while mean filtering caused excessive blurring. Median filtering was especially effective in preserving edges, particularly in text regions, demonstrating its robustness to noise.

In Task 3, frequency-domain filtering provided a more direct interpretation of image structures. Low-pass filtering produced smoothing effects similar to spatial filters but in a more controlled way, while high-pass filtering effectively highlighted edges and fine details. This demonstrated the strong relationship between frequency components and spatial structures.

In Task 4, Sobel and Prewitt operators produced stable and continuous edges, with Sobel showing slightly better performance. The LoG method detected finer details but introduced more noise and fragmented edges, especially in textured regions.

In Task 5, Shi–Tomasi outperformed Harris in terms of reliability and stability of detected corners. While Harris detected more points, Shi–Tomasi produced more meaningful and less noisy interest points.

Finally, in Task 6, projective transformation successfully corrected perspective distortion in a planar object, demonstrating the effectiveness of geometric transformations in recovering real-world structure.

Overall, the experiments highlight that no single method is universally optimal; instead, the choice of method depends on the specific task, such as enhancement, smoothing, feature extraction, or geometric correction.

10. Conclusion

This assignment provided a comprehensive understanding of fundamental low-level vision techniques, including image enhancement, filtering, frequency analysis, edge detection, corner detection, and geometric transformations. The results demonstrated that different methods are suitable for different scenarios: CLAHE was most effective for low-light enhancement, Gaussian filtering provided a good balance between smoothing and detail preservation, Sobel offered stable edge detection, and Shi–Tomasi produced reliable corner detection. Additionally, projective transformation successfully corrected perspective distortion in planar objects. Overall, this pipeline illustrates how classical computer vision techniques can be combined to analyze and improve image quality under various conditions.

11. AI Usage Disclosure

This report and MATLAB implementation were developed by the student with the support of generative AI tools. AI was used for guidance, code refinement, and improving explanations, while all main tasks, experiments, and result evaluations were carried out and verified by the student.

12. Appendix A - Suggested Results Table

Image	Task	Best Method	Key Metric(s)	Short Note
wp2 / wp3 / wp4	Task 1	CLAHE	Contrast improvement (visual)	Best for low-light enhancement
wp1 / wp5 / wp7	Task 2	Gaussian Filter	SSIM (best balance)	Noise vs detail trade-off
wp1 / wp7	Task 3	Low-pass / High-pass	Cutoff radius (qualitative)	Frequency behavior observed
wp1 / wp6 / wp7	Task 4	Sobel	Edge density / continuity	Edge continuity and sensitivity
wp1 / wp5 / wp6 / wp7	Task 5	Shi-Tomasi	Number of corners	Interest point stability
wp6 / wp15	Task 6	Projective Transform	Output width / height	Planar homography